PERSONAL INFORMATION	Name: Adress: Telephone: E-mail: Nationality: Date of birth:	Barış İrhan Gastdozentenhaus Pfaffenwaldring 54, 70569 Stuttgart (+49) 17674519826 barisirhan@gmail.com Turkish 10.04.1977
EDUCATION	➢ Dr.-Ing., Institute of Construction Materials 2011-2014, University of Stuttgart, Stuttgart, Germany Magna cum laude Adviser: Prof. Dr.-Ing. Joško Ožbolt Thesis: "*High velocity impact and fragmentation of concrete: Numerical simulation*" ➢ M.Sc., Computational Mechanics of Materials and Structures (COMMAS) 2001-2003, University of Stuttgart, Stuttgart, Germany GPA: 1.0 (Ranked 1st among 28 students) Adviser: Prof. Dr.-Ing. Christian Miehe Thesis: "*Comparison of alternative kinematical approaches to finite element formulations with embedded strong discontinuities*" ➢ M.S., Civil Engineering 1999-2002, Middle East Technical University, Ankara, Turkey GPA: 3.92/4.00 Adviser: Assoc. Prof. Dr. Uğur Polat Co-adviser: Prof. Dr. Yalçın Mengi Thesis: "*Interaction analysis for the response of structures subjected to seismic excitations: A parametric study*" ➢ B.S., Civil Engineering 1994-1999 Middle East Technical University, Ankara, Turkey GPA: 3.56/4.00 (Ranked 6th among approximately 180 students) ➢ High School 1991-1994 Yuce Science High School, Ankara, Turkey Graduation with distinction ➢ Secondary School 1988-1991 Yahya Kemal Beyatli Secondary School, Ankara, Turkey Graduation with distinction	
PROFESSIONAL EXPERIENCE	➢ Research Associate 2018-2019, Institute of Construction Materials, Stuttgart University, Stuttgart, Germany ➢ Software Engineer 2008-2018, PROYA, Ankara, Turkey ➢ Military Service 2006-2007, MSB İzmir İnşaat Emlak, İzmir, Turkey ➢ Research Assistant 2005-2006, Institute of Construction Materials, Stuttgart University, Stuttgart,	

	Germany ➢ Research Assistant 2003-2004, Institute for Computational Mechanics, Technical University of Munich , Munich, Germany ➢ Teaching Assistant 2000-2001, CE 384 Structural Analysis, Middle East Technical University, Ankara, Turkey
PROFESSIONAL SKILLS	➢ Fortran, C++, Matlab. ➢ Selection and rendering algorithms using OpenGL graphics library. ➢ PROYA Software Company • Worked as a member of software team gathered to develop COMOSYS (Construction Modelling Systems) program, an object oriented software for the analysis, design and detailing of commercial and industrial buildings. • Developed GUI, topological data structures and algorithms for 3D positioning of reinforcement bars. ➢ Finite Element Method • Developer of general purpose finite element program for the simulation of 3D large-deformation frictionless-contact using Matlab. • Developer of CIF (Contact Impact & Fragmentation) 3D explicit finite element analysis program. GUI (Pre/Post) by Borland C++, FE-Core by Intel Fortran. • Developer of CIF-JOB simulation manager program. • Developer of CIF-FEMAP interface for pre- and post- processing (http://www.plm.automation.siemens.com/en_gb/products/femap/) • Developer of really general purpose finite element research and analysis program based on plug-in-play client-server message passing interface. GUI (Pre/Post) by Borland C++, FE-Core by Intel Fortran. • Algorithms for 2D quasi-static cohesive crack growth simulation using finite elements with embedded strong discontinuities based on nodal (XFEM formulation) and elemental (SOS, KOS and SKON formulations) enrichments. • Algorithms for 2D isotropic-damage-model of traction-separation based on internal variable formulation. • Algorithms for 2D two-phase fluid flow simulation based on coupled Levelset-XFEM method. • Algorithms for the stabilization of convection-dominated transport equations, in particular Levelset function, based on SUPG formulation. • Algorithms for 3D implicit large-deformation frictionless-contact based on

	penalty and augmented-lagrangian formulations. • Algorithms for 3D explicit large-deformation frictional-contact with element erosion based on predictor-corrector type lagrange-multiplier-like penalty formulation. • Algorithms based on topology of underlying finite element discretization for (i) contact with element erosion in 3D, and (ii) global tracking of displacement discontinuities in 2D. • Algorithms for volume-coupled transient thermo-mechanical contact simulations based on staggered approach. • Algorithms for large-strain extension of rate and temperature dependent microplane material model for the constitutive modeling of plain-concrete. • Algorithms for adaptive-mesh-refinement, adaptive-remeshing and adaptive-element-deletion in 2D and 3D. • Conducted diploma thesis onChimera overlapping domain-decomposition method for fluid solid interaction.
JOURNAL PUBLICATIONS	➢ B. İrhan , J. Ožbolt, and D. Ruta, 2015. "3D finite element simulations of high velocity projectile impact", *International Journal of Solids and Structures,* 72, pp. 38-49 ➢ N. Bede, J. Ožbolt, A. Sharma and B. İrhan, 2015. "Dynamic fracture of notched plain concrete beams: 3D finite element study", *International Journal of Impact Engineering*, 77, pp.176-188 ➢ J. Ožbolt, A. Sharma, B. İrhan and E. Sola, 2014. "Tensile behavior of concrete under high loading rates", *International Journal of Impact Engineering*, 69, pp.55-68...
SPECIAL SKILLS	➢ Languages: Turkish (native), German (little), English (very good). ➢ Tennis, dance.